

# Computational Supplement to “A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal $\tau$ -adic Wess–Zumino Resolution”

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## Abstract

This supplement records the exact finite computations underlying the strict one-loop BV obstruction and its formal  $\tau$ -adic compensator resolution. It separates basis exhaustiveness, relative-cohomology ranks, antifield and gauge-fixing transport, analytic coefficient provenance, the compensator cotangent lift, quartet contraction, the extended quotient, and QME restoration. Tables and matrices carrying numerical or discrete claims are generated directly from content-addressed JSON certificates. Independent verifiers reject stale generated material, altered ranks, missing classes, over-promoted lifecycle states, and dependency-hash drift.

## Scope and dependency labels

The supplement proves only the claims labeled `LOCAL-ALGEBRAIC` and `EUCLIDEAN-SPECTRAL` in the companion manuscript. It does not prove a Lorentzian QME, renormalized causal products, a Hadamard state, positivity, particles, residual transfer, or an all-loop theorem. The strict and compensator-extended theories remain different rows throughout.

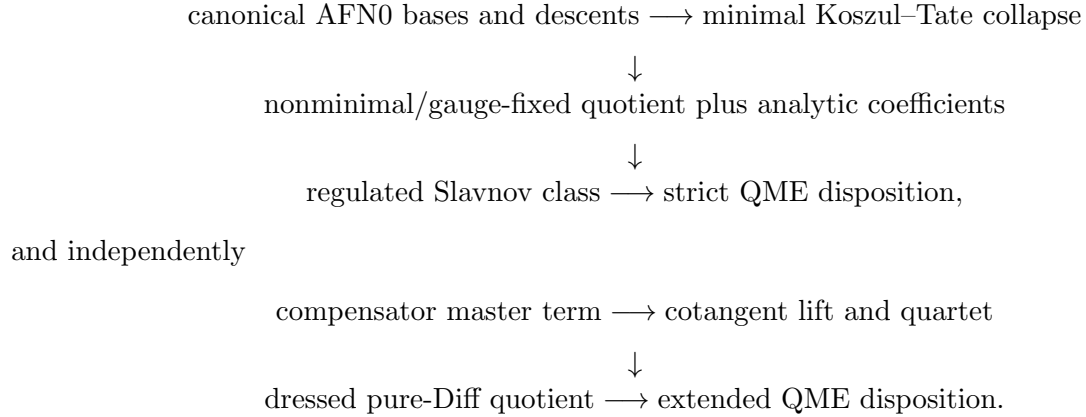
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## 1 Certificate graph

The calculation is organized as a directed acyclic graph rather than a single symbolic notebook:



No determinant row is promoted directly into BV cohomology. The analytic coefficient vector is accepted only after it is reduced against the complete gauge-fixed quotient with the same normalization and classical snapshot.

## 2 Exact strict quotient

### 2.1 Declared local algebra

The production slice is four-dimensional, form degree four, engineering dimension four, and ghost number zero or one. Tensor generators include the metric, curvature and covariant derivatives, the Diff and Weyl ghosts, the minimal antifields, and the declared nonminimal doublets. Epsilon tensors are included in the odd sector. The canonical quotient imposes:

- algebraic and differential Bianchi identities;
- tensor symmetries and four-dimensional antisymmetrization;
- graded factor permutations and Grassmann signs;
- dummy-index graph isomorphism;

- integration by parts and horizontal exactness.

The basis generator is checked twice: forward tensor-graph generation and reverse coverage of every grading-admissible signature. A class receives a complete nontriviality witness only after the top, lower-form cocycle, and lower-form boundary spaces are exhaustive.

## 2.2 Orbit-first reduction

The even ghost-one basis has seven canonical top carriers before closure and boundary reduction. Orbit-first contraction resolves the two formerly pending mixed signatures by materializing only thirty raw pairings. The odd calculation resolves all three pending mixed signatures: forty-five raw graphs reduce to nine signed orbits, all of which vanish by the Bianchi relations. The single odd Weyl pseudoscalar remains.

The exact rank data and final dimensions are generated below.

| Stage                    | $\dim H_{\text{even}}^{0,4}$ | $\dim H_{\text{odd}}^{0,4}$ | $\dim H_{\text{even}}^{1,4}$ | $\dim H_{\text{odd}}^{1,4}$ |
|--------------------------|------------------------------|-----------------------------|------------------------------|-----------------------------|
| strict gauge-fixed BV    | 2                            | 1                           | 2                            | 1                           |
| $\tau$ -adic extended BV | 3                            | 1                           | 0                            | 0                           |

Table 1: Exact dimension-four local BV quotient dimensions.

| Spectral-sequence step | Exact input                           | Rank/count      | Outcome                     |
|------------------------|---------------------------------------|-----------------|-----------------------------|
| Koszul–Tate $E_0$      | adapted regular-Bach pairs            | 6 pairs         | positive AFN columns vanish |
| $E_1 \Rightarrow E_2$  | AFN0 modulo Euler–Noether ideal       | 9016 monomials  | collapse at $E_2$           |
| nonminimal extension   | pointwise Diff $\times$ Weyl doublets | 10 pairs        | chain contraction           |
| gauge-fixing transport | local BV-canonical similarity         | 25080 monomials | quotient preserved          |

Table 2: Strict-quotient spectral-sequence and contraction ledger on the regular Bach locus.

| Sector                     | raw graphs/pairings materialized | closure rank | boundary rank |
|----------------------------|----------------------------------|--------------|---------------|
| even AFN0 relative carrier | 30                               | 6            | 4             |
| odd AFN0 mixed carrier     | 45                               | 1            | 0             |

Table 3: Orbit-first strict AFN0 quotient audit. The ambient  $2.86 \times 10^9$  raw-graph expansion is never materialized.

$$Q_W = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad Q_W h + h Q_W = I_4, \quad (1)$$

in the ordered basis  $(\tau, \omega, \omega^*, \hat{\tau}^*)$ . The certificate checks 24 atoms and 8 generators over  $Q$ ; all component gradings,  $\delta^2$ ,  $\delta\gamma + \gamma\delta$ , and  $Q^2$  pass exactly.

| Sector                 | operator                      | rank | statistics | exponent in $Z$ | zero modes removed | local prescription                        |
|------------------------|-------------------------------|------|------------|-----------------|--------------------|-------------------------------------------|
| metric TT, upper       | <code>Delta_2_perp(4)</code>  | 5    | bosonic    | $-\frac{1}{2}$  | 0                  | factorwise spectral cut                   |
| Diff–Weyl scalar ghost | <code>Delta_0(-4)</code>      | 1    | fermionic  | $\frac{1}{2}$   | 5                  | cut; negative-mode phase locally constant |
| metric TT, lower       | <code>Delta_2_perp(2)</code>  | 5    | bosonic    | $-\frac{1}{2}$  | 0                  | factorwise spectral cut                   |
| transverse Diff ghost  | <code>Delta_1_perp(-3)</code> | 3    | fermionic  | $\frac{1}{2}$   | 10                 | factorwise spectral cut                   |

Table 4: Human-readable determinant ledger for the accepted repository Euclidean integration slice. Exponents are determinant powers in  $Z$ ; the corresponding  $\Gamma_1$  powers have the opposite sign.

| Sector                 | $C^2$            | $E_4$              | $C\tilde{C}$ | $\square R$ |
|------------------------|------------------|--------------------|--------------|-------------|
| metric TT, upper       | $\frac{37}{24}$  | $-\frac{31}{72}$   | 0            | 0           |
| Diff–Weyl scalar ghost | $\frac{89}{120}$ | $-\frac{269}{360}$ | 0            | 0           |
| metric TT, lower       | $\frac{29}{8}$   | $-\frac{181}{72}$  | 0            | 0           |
| transverse Diff ghost  | $\frac{29}{40}$  | $-\frac{79}{120}$  | 0            | 0           |
| exact sum              | $\frac{199}{30}$ | $-\frac{87}{20}$   | 0            | 0           |

Table 5: Factorwise local heat-kernel coordinates after the accepted determinant exponents, measure and quartet cancellations. Their exact sum is the bosonic-Weyl-parameter Ward carrier before ghost replacement and BV quotient reduction.

The Euler primitive used by the boundary matrix is

$$B_E = \int \sqrt{g} \left[ \tau E_4 + 4G^{\mu\nu} \nabla_\mu \tau \nabla_\nu \tau - 4(\square \tau)(\nabla \tau)^2 + 2(\nabla \tau)^4 \right], \quad Q_W B_E = \int \sqrt{g} \omega E_4 \pmod{d_h}, \quad (2)$$

with the sign convention  $Q_W B_C = \int \sqrt{g} \omega C^2$  and  $Q_W B_E = \int \sqrt{g} \omega E_4$  modulo  $d_h$ .

$$B_{\text{ext}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{rank } B_{\text{ext}} = 4, \quad (3)$$

from the ordered primitive basis  $(B_C, B_E, B_P, B_\square)$  to  $(\omega C^2, \omega E_4, \omega C\tilde{C}, \omega \square R)$ . The strict breaking and its extended boundary image are both

$$\left( \frac{199}{30}, -\frac{87}{20}, 0, 0 \right).$$

| Density coordinate | repository one-loop coefficient |
|--------------------|---------------------------------|
| $C^2$              | $\frac{199}{30}$                |
| $E_4$              | $-\frac{87}{20}$                |
| $C\tilde{C}$       | 0                               |
| $\square R$        | 0                               |

Table 6: Coefficient vector in the convention  $(4\pi)^{-2}[cC^2 - aE_4 + pC\tilde{C} + b\square R]$ .

| Standard-sign field             | $C^2$ coordinate | $E_4$ coordinate   |
|---------------------------------|------------------|--------------------|
| pure Weyl graviton              | $\frac{199}{30}$ | $-\frac{87}{20}$   |
| real conformal scalar           | $\frac{1}{120}$  | $-\frac{360}{11}$  |
| Weyl fermion                    | $\frac{40}{1}$   | $-\frac{720}{11}$  |
| Dirac fermion                   | $\frac{1}{20}$   | $-\frac{360}{31}$  |
| gauge vector (with BRST ghosts) | $\frac{1}{10}$   | $-\frac{180}{180}$ |

Table 7: Exact anomaly vectors in the convention  $(c, -a)$ . The functional  $(1, 0)$  is strictly positive on the gravity vector and every allowed matter ray, already excluding cancellation in the nonnegative cone.

## 2.3 Antifields and gauge fixing

The imported minimal classical export supplies every Koszul–Tate row. On the regular Bach locus the positive-antifield sector is an explicit sum of adapted pairs, and its contracting homotopy verifies

$$\delta\sigma + \sigma\delta = N_{\text{pair}}.$$

Consequently the AFN0 Weyl representatives lift through the minimal spectral sequence. A separate total-complex computation proves that the pure-Diff and mixed Diff–Weyl sectors add no independent dimension-four class.

Concretely, the six adapted pairs are  $(g^*, E_g)$ ,  $(\xi^*, N_\xi)$ ,  $(\omega^*, N_\omega)$  and their Lie-prolonged partners. The  $E_0$  positive-antifield columns vanish,  $E_1$  is AFN0 modulo the Euler–Noether ideal, and the sequence collapses at  $E_2$ . Since every positive-antifield generator in this slice has form degree four, no omitted lower-form positive-AFN carrier can alter the relative quotient. This is a chart theorem: replacing highest TT metric jets by the Bach Euler row assumes the declared regular Bach locus.

Ten general nonminimal pairs then contract on 25,080 graded regression monomials. Their covariant jet prolongations commute with  $d_h$ , and a free-word normal-form computation transports the SDR through an arbitrary invertible local BV-canonical gauge-fixing transformation. Thus the minimal, nonminimal, and gauge-fixed quotient dimensions agree. This explicit chain comparison is why the main theorem is not labeled AFN0-only.

## 3 Coefficient and Slavnov route

### 3.1 Analytic input

The accepted Euclidean principal-symbol sequence is

$$(T \oplus \mathbf{1})_5 \xrightarrow{\sigma(K)} (S^2 T^*)_5 \xrightarrow{\sigma(P_{\text{TT}})} (S_{\text{TT}}^2 T^*)_5.$$

| Anomaly-induced functional carrier                 | exact coefficient |
|----------------------------------------------------|-------------------|
| $\langle \mathcal{E}_4, G_4 C^2 \rangle$           | $\frac{199}{120}$ |
| $\langle \mathcal{E}_4, G_4 \mathcal{E}_4 \rangle$ | $-\frac{87}{160}$ |
| $\int \sqrt{g} R^2$                                | $\frac{29}{120}$  |

Table 8: Exact Paneitz/Riegert solve. The Weyl-response matrix is  $\text{diag}(4, 8, -12)$ ; re-expansion returns  $(199/30, -87/20, 0)$  in  $(C^2, E_4, \square R)$ .

| Flat-TT form-factor datum   | exact value       |
|-----------------------------|-------------------|
| $c$                         | $\frac{199}{30}$  |
| $\beta_2$                   | $\frac{199}{15}$  |
| $A_{\log}$                  | $-\frac{199}{60}$ |
| $\mu \partial_\mu F_C$      | $\frac{199}{30}$  |
| finite local $C^2$ constant | open              |

Table 9: Universal nonzero-momentum flat Euclidean TT logarithmic form factor. The additive finite normalization and curved completion are not fixed.

The first bundle contains four Diff plus one Weyl gauge parameter; the middle bundle is the metric perturbation; the last is the TT symbol slice, not a five-component claim about the unreduced Bach equation. It has ranks  $(5, 5)$  at nonzero covector, zero composition, and exact image–kernel equality. The formal adjoint sequence, Diff/Weyl nonminimal doublets, and four kinetic blocks replay independently. The determinant bundle ranks and exponents are

$$(5, 1, 5, 3), \quad \left(-\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}\right),$$

with priming dimensions  $(0, 5, 0, 10)$ . The generated determinant table prints each operator, rank, statistics, exponent, and zero-mode subtraction separately.

For each factor, the two invariant coordinates are reconstructed from independent carriers: the Ricci-flat row supplies  $c_i - a_i$  and the round- $S^4$  row supplies  $a_i$ , so the generated table forms  $c_i = (c_i - a_i) + a_i$  before summing factors. The parity-odd zero is factorwise for the declared real parity-even operator family. The  $\square R = 0$  column is instead the chosen scheme for the total functional, implemented by one local  $R^2$  counterterm; no factorwise scheme-invariant type-D claim is made.

The local  $C^2$  coordinate is exposed on a non-conformally-flat Ricci-flat carrier. The Euler coordinate is independently fixed on round  $S^4$ . The measure ledger includes York and ghost super-Jacobians, all nonminimal quartets, conformal-Killing zero modes, the oriented auxiliary contour, and the local parity Ward identity. Global determinant phase and conformal-group volume are not used to infer a local coefficient.

### 3.2 Cohomological reduction

Let the ordered strict basis be

$$(\omega C^2, \omega E_4, \omega C\tilde{C}, \omega \square R).$$

The raw analytic vector is the one displayed in Table 6. The fourth coordinate is removed with the stored  $R^2$  primitive and current. The first two normalized dual functionals evaluate to  $199/30$  and

| Covariant-log datum                                           | exact value       |
|---------------------------------------------------------------|-------------------|
| curvature order of $\langle C, \log(\Delta_C/\mu^2)C \rangle$ | 2                 |
| first admissible operator-choice difference                   | 3                 |
| logarithmic coefficient                                       | $-\frac{199}{60}$ |
| selected FV Weyl-orbit completion                             | certified         |
| first forced FV correction order                              | 3                 |
| independent cubic Weyl-invariant form factors                 | open              |
| finite $C^2/R^2$ normalization                                | open              |

Table 10: Covariant curvature-squared logarithm, its exact FV completion as a selected carrier, and the independent-data boundary. The coefficient remains  $-\frac{199}{60}$ .

$-87/20$ , while the odd functional evaluates to zero by the Ward identity. Therefore

$$[\mathcal{A}^{(1)}] = \frac{1}{(4\pi)^2} \left( \frac{199}{30} [\omega C^2] - \frac{87}{20} [\omega E_4] \right) \neq 0.$$

The strict QME verdict is determined by this nonzero cohomology class. The beta function, determinant logarithm, and background trace anomaly are not used as substitutes for the regulated Slavnov insertion.

The bridge is the regulated identity

$$\mathcal{S}_S \Gamma_{\text{reg}}^{(1)} = \lim_{d \rightarrow 4} s_d \left[ \frac{\mu^{d-4}}{(4\pi)^2(d-4)} \int \sqrt{g} (cC^2 - aE_4 + b\Box R + \text{exact rows}) \right].$$

The  $(d-4)$  Weyl variation of the continued densities cancels the pole. Replacing the bosonic Weyl parameter by  $\omega$ , adjoining the certified universal Diff descent, and reducing exact, parameter-dependent, and total derivative rows produces the ghost-one insertion. Thus the determinant ledger supplies coefficients to an independently assembled Slavnov carrier; it is not itself called a BV anomaly.

In certificate notation the residue map is

$$s_d I_i(d) = (d-4)A_i^4 + s_d B_i(d) + d_h C_i(d) + O((d-4)^2),$$

with  $A_i^4$  the complete ghost-one carrier. The pole residue is paired with the normalized quotient duals only after the exact and horizontal rows are removed. This is the machine-readable distinction among determinant coefficients, a bosonic Weyl Ward variation, and the BV Slavnov class.

### 3.3 Matter cone

For standard-sign free conformal scalars, Weyl/Dirac fermions, and gauge vectors, cancellation is an exact rational cone-feasibility problem. A separating dual covector is positive on every allowed matter ray and on the pure-gravity breaking. Hence no nonnegative real multiplicity vector, and therefore no nonnegative integral multiplicity vector, cancels both even coordinates. This theorem deliberately excludes nonunitary signs, interacting fixed points, conformal higher spin, and compensators.

| Certificate                                                                                                                               | SHA-256 prefix   |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| quantum-weyl/local_bv/certificates/AFNO_H14_EVEN_CANONICAL_QUOTIENT.json                                                                  | 6bf8b7ef099ce40b |
| quantum-weyl/local_bv/certificates/AFNO_H14_ODD_CANONICAL_QUOTIENT.json                                                                   | 0134e28cf3fd0bf4 |
| quantum-weyl/local_bv/certificates/GENERAL_NONMINIMAL_GAUGE_FIXED_CONTRACTION.json                                                        | 4513be4824760577 |
| quantum-weyl/local_bv/certificates/MINIMAL_BV_KOSZUL_TATE_COLLAPSE.json                                                                   | f2b04706724d79b5 |
| quantum-weyl/spectral/euclidean/certificates/REPOSITORY_EUCLIDEAN_ELLIPTIC_COMPLEX.json                                                   | a1eb7dd598ed3417 |
| quantum-weyl/spectral/euclidean/certificates/REPOSITORY_FULL_BV_MULTIPLICITY_LEDGER.json                                                  | 5bdc6ee4674e3a58 |
| quantum-weyl/spectral/euclidean/certificates/STANDARD_EUCLIDEAN_LOCAL_B4_INTEGRATION_SLICE.json                                           | edde6206a9dc62d0 |
| quantum-weyl/spectral/euclidean/certificates/REPOSITORY_NONCONFORMALLY_FLAT_OR_RICCI_FLAT_FULL_BV_OPERATOR_MEASURE_COEFFICIENT_MATCH.json | d23543cdab828288 |
| quantum-weyl/anomalies/certificates/REGULATED_REPOSITORY_BV_SLAVNOV_BREAKING.json                                                         | d11ebc782c6a388e |
| quantum-weyl/anomalies/certificates/UNITARY_CONFORMAL_MATTER_CANCELLATION_NO_GO.json                                                      | 9c4b1218450e20f0 |
| quantum-weyl/anomalies/certificates/WESS_ZUMINO_MINIMAL_BV_COTANGENT_LIFT.json                                                            | b265dc9d86938ed7 |
| quantum-weyl/anomalies/certificates/WESS_ZUMINO_COMPENSATOR_EXTENSION_PREFLIGHT.json                                                      | 10dbd0b0a1e9190d |
| quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.json                                                         | fa21fc6071ae5227 |
| quantum-weyl/transfer/certificates/ANOMALY_INDUCED_NONLOCAL_GAMMA1.json                                                                   | 9531ef49d0d930fc |
| quantum-weyl/transfer/certificates/FLAT_TT_LOGARITHMIC_GAMMA1.json                                                                        | 8d4bc8bdf702cf3c |
| quantum-weyl/transfer/certificates/CURVATURE_SQUARED_COVARIANT_LOG_GAMMA1.json                                                            | a67a475da6352ad1 |
| quantum-weyl/transfer/certificates/FV_CONFORMIZED_C2_LOG_GAMMA1.json                                                                      | 7a4e55efa7284d30 |

Table 11: Content-addressed inputs to the generated supplement tables. Full hashes are stored in the Paper 12 claim map.

## 4 Compensator BV extension

### 4.1 Rows derived from one master term

The extension is generated from

$$S_{\tau, \min} = \int \tau^* (\xi^\mu \nabla_\mu \tau + \omega).$$

Its four Euler derivatives force

$$\begin{aligned} Q\tau &= \mathcal{L}_\xi \tau + \omega, \\ \delta\omega^* &= N_\omega + \tau^*, \\ \delta\xi_\mu^* &= N_{\xi, \mu} + \tau^* \nabla_\mu \tau, \\ \gamma\tau^* &= \mathcal{L}_\xi \tau^*. \end{aligned}$$

The Lie-prolonged cotangent rows carry the opposite formal-adjoint signs. No coefficient is fitted to a nilpotency defect.



On all twenty-four strict-plus-compensator atoms the independent replay finds  $\delta^2 = 0$ ,  $\delta\gamma + \gamma\delta = 0$ , and  $Q^2 = 0$ . It also checks every ghost number, antifield number, parity, and tensor component type.

## 4.2 Dressed canonical coordinates

The change

$$\hat{g} = e^{-2\tau} g, \quad \hat{g}^* = e^{2\tau} g^*, \quad qquad \hat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu}$$

obeys the canonical one-form identity

$$g^* \delta g + \tau^* \delta \tau = \hat{g}^* \delta \hat{g} + \hat{\tau}^* \delta \tau.$$

It sends the Weyl Noether row to  $\delta\omega^* = \hat{\tau}^*$  and yields the quartet matrices in Equation 1.

Equivalently,  $h\omega = \tau$ ,  $h\hat{\tau}^* = \omega^*$  and  $h$  vanishes on  $\tau, \omega^*$ . Extending  $h$  to covariant jets as a graded derivation and dividing by positive quartet number gives the contraction  $Q_W h + h Q_W = N_{\text{quartet}}$ . Diff transport and  $d_h$  preserve the filtration, so homological perturbation lifts this contraction to the total formal complex.

More directly, write the total differential as  $\mathcal{D} = Q_W + \delta$ . In dressed covariant jets the quartet members occur in matched Diff modules, so the covariant graded derivation  $h$  obeys  $[\delta, h]_+ = 0$  and  $[N_{\text{quartet}}, \delta] = 0$ . Thus

$$\mathcal{D}h + h\mathcal{D} = N_{\text{quartet}}.$$

Every closed component of positive quartet number  $n$  is explicitly the boundary  $\mathcal{D}(n^{-1}h\mathbf{a}_n)$ . At quartet number zero the total differential is the pure-Diff BV differential of  $\hat{g}$ . The independently certified pure-Diff ghost-one zero, rather than the four primitive matrix alone, therefore completes the proof of  $H_{\text{ext}}^{1,4} = 0$ .

The coordinate change is interpreted in the formal  $\tau$ -adic local analytic completion. Exact inverse series are checked through order eight, and the all-orders coefficient convolution is

$$\sum_{k=0}^n \frac{(-2)^k}{k!} \frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!} (1-1)^n = \delta_{n0}.$$

## 5 Extended quotient and restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in  $\hat{g}$ . The dimension-four ghost-zero basis therefore contains the new invariant  $R(\hat{g})^2$  in addition to  $C(\hat{g})^2$ ,  $E_4$ , and the odd Pontryagin density.  $\square R(\hat{g})$  remains horizontally exact.

The only dimension-zero natural symmetric tensor that could generate an incoming positive-antifield boundary is  $\hat{g}$  itself. Its image is the trace of the Bach Euler row and vanishes. Normalized coordinate duals for all four surviving ghost-zero classes annihilate both the horizontal and positive-antifield boundary spaces.

At ghost number one the primitives  $(B_C, B_E, B_P, B_\square)$  map to the complete strict four-candidate carrier by the identity matrix in Equation 3. The independently certified pure-Diff anomaly quotient is zero. Hence the full extended dimension-four  $H^{1,4}$  vanishes in both parities.

The coefficient-bearing counterterm is

$$S_{\text{ct}}^{(1)} = -\frac{\hbar}{(4\pi)^2} \left( \frac{199}{30} B_C - \frac{87}{20} B_E \right),$$

and exact coordinate replay gives

$$\mathcal{A}^{(1)} + s_{\text{ext}} S_{\text{ct}}^{(1)} = 0 \pmod{\mathfrak{d}_h}.$$

This is the one-loop local Euclidean QME restoration theorem in the declared formal  $\tau$ -adic theory. It changes the theory: setting  $\tau = 0$  does not make the strict quotient anomaly-free, and the dressed algebra contains the new  $R(\hat{g})^2$  counterterm direction.

## 6 First Slavnov operator: fixed part and exact ambiguity

The counterterm fixes the local Hamiltonian contribution

$$Q_1^{\text{WZ}} = (C_1^{\text{WZ}}, \cdot).$$

Here

$$C_1^{\text{WZ}} = -(4\pi)^{-2} \left( \frac{199}{30} B_C - \frac{87}{20} B_E \right).$$

Since  $C_1^{\text{WZ}}$  has antifield number zero, its field and ghost rows vanish and its cotangent rows are the Euler derivatives with respect to  $\hat{g}$  and  $\tau$ . This is an actual coefficient-bearing part of  $Q_1$ , but not the complete renormalized operator.

The exact non-uniqueness witness uses flat Euclidean momentum  $p = (1, 0, 0, 0)$ , a TT polarization  $h_{11} = 1, h_{22} = -1$ , and the conformal polarization  $h_{\mu\nu} = \delta_{\mu\nu}$ . Direct contraction of the linearized Riemann tensor gives

|           | $C^{(1)2}$ | $R^{(1)2}$ |
|-----------|------------|------------|
| TT        | 1          | 0          |
| conformal | 0          | 9          |

Including the topological Euler and Pontryagin columns, the allowed extended  $H^{0,4}$  response matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}, \quad \text{rank} = 2.$$

The topological columns have zero compactly supported bulk Euler derivative; their boundary and global content is retained separately. Therefore finite  $C(\hat{g})^2$  and  $R(\hat{g})^2$  counterterms change the bulk Hamiltonian vector field independently. This receipt alone does not supply a nonlocal  $\Gamma_1$  representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The follow-up below supplies the anomaly-induced component but not the Weyl-invariant finite remainder. Complete  $Q_1$  remains fail-closed as `NO_CERTIFIED_OPERATOR`.

### 6.1 Anomaly-induced nonlocal representative

The follow-up receipt solves the exact Weyl-response system rather than declaring the whole finite effective action absent. With

$$\mathcal{E}_4 = E_4 - \frac{2}{3} \square R, \quad \Delta_4 = \square^2 + 2R^{\mu\nu} \nabla_\mu \nabla_\nu - \frac{2}{3} R \square + \frac{1}{3} (\nabla^\mu R) \nabla_\mu,$$

the carrier basis

$$\left( \langle \mathcal{E}_4, G_4 C^2 \rangle, \langle \mathcal{E}_4, G_4 \mathcal{E}_4 \rangle, \int \sqrt{g} R^2 \right)$$

has Weyl-response matrix  $\text{diag}(4, 8, -12)$ . Exact rational elimination against the modified-basis target  $(199/30, -87/20, -29/10)$  gives

$$\left(\frac{199}{120}, -\frac{87}{160}, \frac{29}{120}\right).$$

Re-expansion in  $(C^2, E_4, \square R)$  returns  $(199/30, -87/20, 0)$  exactly. Thus

$$\Gamma_{1,\text{anom}} = \frac{1}{(4\pi)^2} \left\{ \frac{1}{8} \langle \mathcal{E}_4, G_4(2cC^2 - a\mathcal{E}_4) \rangle + \frac{a}{18} \int \sqrt{g} R^2 \right\}$$

is one anomaly-induced Euclidean representative.

The inverse contract is fail-closed. Either the Euclidean boundary problem is invertible, or  $G_4$  is a self-adjoint generalized inverse and every displayed source lies in the complement of  $\ker \Delta_4$ . A compact Paneitz zero mode is not silently discarded: its global anomaly, homogeneous solutions, and boundary transgressions remain separate data. The Weyl-invariant finite functional, finite  $C^2$  and absolute dressed  $R(\hat{g})^2$  normalization, and renormalized BV product are still absent, so this result does not promote complete  $\Gamma_1$  or  $Q_1$ .

## 6.2 Universal nonzero-momentum flat-TT logarithm

The conformal-sector receipt next fixes the part left invariant by the inhomogeneous anomaly solve at quadratic order on a declared flat Euclidean TT carrier. With the fixture normalized by  $\langle C^{(1)}, C^{(1)} \rangle = 1$ , exact rational arithmetic gives

$$F_C(p^2; \mu) = -\frac{199}{60} \log \frac{p^2}{\mu^2} + z_C(\mu), \quad -\frac{199}{60} = -\frac{c}{2} = -\frac{\beta_2}{4}.$$

The independent checks replay

$$\mu \partial_\mu F_C = \frac{199}{30}, \quad F_C(p^2) - F_C(q^2) = -\frac{199}{60} \log \frac{p^2}{q^2},$$

and verify that the prior Riegert representative begins at  $O(h^4)$  on this pure-TT slice. A finite reference condition only determines the family

$$z_C(\mu) = N_C + \frac{199}{60} \log \frac{\kappa_{\text{ref}}^2}{\mu^2};$$

neither  $N_C$  nor  $\kappa_{\text{ref}}$  is selected by the anomaly. The receipt therefore rejects promotion to a curved completion, a zero-momentum result, a Lorentzian branch prescription, complete  $\Gamma_1/Q_1$ , or a particle entry.

## 6.3 Curvature-squared covariantization

The next receipt replaces the flat momentum logarithm by the spectral logarithm of a positive self-adjoint Laplace-type operator  $\Delta_C$  on the Weyl-tensor bundle. On the complement of its kernel,

$$\Gamma_{1,C^2 \log}^{(2)} = -\frac{1}{(4\pi)^2} \frac{199}{60} \langle C, \Pi_\perp \log(\Delta_C/\mu^2) \Pi_\perp C \rangle.$$

The verifier replays the coefficient and scale response and applies the curvature filtration to the resolvent Fréchet derivative. For  $\Delta'_C = \Delta_C + V_1 + O(\mathcal{R}^2)$ ,

$$d \log(\Delta_C)[V_1] = \int_0^\infty (\Delta_C + s)^{-1} V_1 (\Delta_C + s)^{-1} ds,$$

so the sandwiched change has exact order  $1 + 1 + 1 = 3$ . Operator-choice dependence for this  $C^2$  carrier therefore starts with cubic form factors within the class sharing principal symbol, connection, domain, and source complement and differing at first curvature order only by the self-adjoint endomorphism  $V_1$ . Connection variations, which can introduce first-order differential terms, are not covered. Kernel and boundary data, independent cubic form factors, finite  $C^2$  constant and absolute dressed  $R(\widehat{g})^2$  normalization remain fail-closed. The selected carrier's Weyl-orbit completion is handled separately below.

#### 6.4 Fradkin–Vilkovisky completion of the selected carrier

The conformization receipt constructs

$$L_g = \square_g - \frac{1}{6}R_g, \quad u_g = 1 + \frac{1}{6}L_g^{-1}R_g, \quad \bar{g}[g] = u_g^2 g.$$

It independently checks  $L_g u_g = 0$ ,  $R[\bar{g}] = 0$ , the Weyl-weight ledger  $2(-1) + 2 = 0$ , and the boundary-normalized transformation  $u_{e^{2\sigma}g} = e^{-\sigma} u_g$ . Consequently  $\bar{g}[e^{2\sigma}g] = \bar{g}[g]$ , and evaluation of the full selected  $C \log \Delta_C C$  carrier—its tensors, pairing, projector, domain and logarithm—on  $\bar{g}$  gives an exact nonlocal functional invariant under the declared local Weyl transformations.

The verifier retains the inverse-domain hypothesis, replays the coefficient  $-199/60$  and the first correction order three, and rejects promotion to independent cubic form factors, complete  $\Gamma_1/Q_1$ , residual transfer or a Lorentzian result. It also enforces a carrier crosswalk: the nonlocal  $\bar{g}[g]$  is distinct from the local  $\tau$ -adic field  $\widehat{g}[g, \tau]$ .

#### 6.5 FV anomaly action and dependent Ricci sector

The coefficient-bearing anomaly action is replayed as

$$\Gamma_{1,\text{FV}} = \frac{1}{(4\pi)^2} \int \sqrt{g} \left[ \left( \frac{199}{30} C^2 - \frac{87}{20} \mathcal{E}_4 \right) \Sigma_{\text{FV}} + \frac{87}{10} \Sigma_{\text{FV}} \Delta_4 \Sigma_{\text{FV}} + \frac{29}{120} R^2 \right], \quad \mathcal{E}_4 = E_4 - \frac{2}{3} \square R.$$

The independent verifier checks both exact cancellations

$$-\frac{87}{5} + \frac{87}{5} = 0, \quad \frac{29}{10} - \frac{29}{10} = 0,$$

and the RFT specialization  $(199/120, -87/160, 29/120)$ .

The decomposition  $\Gamma_1[g] = \Gamma_{1,\text{FV}}[g] + W_1[\bar{g}[g]]$  with  $R[\bar{g}] = 0$  makes the Ricci-scalar sector dependent. A generic-basis term  $RF(\square)R$  can appear after re-expansion, but its form factor is fixed by the anomaly action and the conformized Weyl sector. It is not a separate coefficient gate in the declared massless conformal FV scope. Independent cubic-and-higher Weyl form factors, finite normalizations, global inverse data and renormalized products remain open.

## 6.6 Exact local BoxR scheme conversion

The zeta/proper-time calculation imports the published local heat-kernel rows for the traceless tensor, gauge-weight and diffeomorphism-ghost operators. An independent exact replay of their  $(2, -1, -2)$  determinant combination gives

$$b_{\square}^{\text{raw}} = \frac{7}{2} \log \frac{3}{2} - \frac{159}{80} < 0.$$

The repository primitive  $\omega \square R = -(1/12)s(R^2)$  modulo  $d$  then fixes

$$z_R^{\text{raw} \rightarrow 0} = \frac{7}{24} \log \frac{3}{2} - \frac{53}{320}.$$

The verifier checks that  $b_{\square}^{\text{raw}} - 12z_R^{\text{raw} \rightarrow 0} = 0$  and that adding the same shift to  $391/960 - (7/24) \log(3/2)$  gives the stored local coefficient  $29/120$ . Alternating-series rational bounds certify the strict sign without floating-point arithmetic. The receipt keeps the strict-metric local scheme counterterm separate from the independent dressed  $R(\hat{g})^2$  class in the compensator theory. The subsequent FV certificate, not the local scheme conversion alone, proves that generic-basis nonlocal Ricci terms are structurally determined.

The same certificate binds the contraction non-merge: the Berger ‘tau’ row is the temporal diffeomorphism ghost in the positive-Berger clock complex, whereas the Wess–Zumino ‘tau’ row obeys  $Q\tau = \mathcal{L}_{\xi}\tau + \omega$ . The identical printed symbol is not a cross-background carrier map.

## 7 Independent verifiers and mutation tests

The producer and verifier paths are distinct. The verifiers independently:

- validate strict Draft 2020–12 schemas;
- replay exact matrix ranks and contraction identities;
- compare generated certificates with fresh deterministic builds;
- reject deletion of the dressed  $R^2$  class;
- reject authorization of residual transfer;
- reject promotion of the fixed Wess–Zumino contribution to a complete renormalized  $Q_1$ ;
- replay the diagonal Paneitz/Riegert response solve and reject promotion of a generalized inverse to an unconditional global inverse;
- replay the flat-TT logarithmic coefficient, scale response and momentum difference, and reject promotion of its additive constant or curved completion;
- replay the curvature-order filtration for the covariant logarithm and reject promotion of the cubic remainder, finite constants or Berger/WZ contraction merge;
- replay the Fradkin–Vilkovisky scalar-flat and Weyl-weight identities, and reject identification of its nonlocal representative with the local  $\tau$ -adic dressed metric or promotion to independent cubic form factors;

- replay the exact FV anomaly-action response cancellations and RFT specialization, while rejecting a separate nonlocal  $R^2$  datum or a complete  $\Gamma_1$ ;
- replay the raw  $\square R$  heat-kernel arithmetic, its exact strict-metric  $R^2$  scheme shift and the 29/120 cross-check, while rejecting promotion to an absolute dressed normalization;
- reject changes to the cotangent signs, gradings, or quartet identity;
- validate every content hash used by the paper claim map.

The generated tables add a publication-layer drift guard. Their verifier loads all authoritative table inputs, checks the exact extended boundary matrix and coefficient image, rebuilds the TeX byte-for-byte, and rejects a stale include.

## 8 Reproduction commands

Run from the repository root:

```
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_minimal_bv_cotangent_lift --check
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_extended_local_bv_cohomology --check
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
  transfer.one_loop_slavnov_q1_disposition --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_one_loop_slavnov_q1_disposition
PYTHONPATH=quantum-weyl python3 -m \
  transfer.flat_tt_logarithmic_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_flat_tt_logarithmic_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.curvature_squared_covariant_log_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_curvature_squared_covariant_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.fv_conformized_c2_log_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_conformized_c2_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.fv_anomaly_action_ricci_sector --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_anomaly_action_ricci_sector
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_box_r_scheme_conversion
python3 paper/generate_12_quantum_anomaly_tables.py --check
python3 paper/verify_12_quantum_anomaly_tables.py
python3 paper/generate_12_pure_weyl_one_loop_bv_anomaly_claim_map.py --check
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py
```

The intentionally slow repository-wide suite is not required for this scoped artifact. The relevant exact unit modules, independent verifiers, atlas validator, and two-pass LaTeX build form the declared publication tier.

## 9 Remaining gate

The next theorem requires actual new physical input:

1. a same-background compensator-inclusive classical contraction  $(\iota_{\text{ext}}, \pi_{\text{ext}}, S_{\text{ext}})$ ;
2. the independent cubic-and-higher Weyl-invariant form factors, finite  $C^2$  and absolute dressed  $R(\widehat{g})^2$  normalization, global Paneitz Green data, and renormalized BV operator data fixing a complete  $Q_1$  beyond its certified local Wess–Zumino, anomaly-induced, and flat-TT logarithmic contributions;
3. the induced residual operator  $q_1 = \pi_{\text{ext}} Q_1 \iota_{\text{ext}}$ ;
4. nilpotency, adjacent-cohomology, Cartan, mixing, and pairing checks.

Until these land, residual transfer is forbidden. Same-background free-state and Hadamard constructions belong to a separate companion programme and are not claims of this paper. The classical-obstruction-to-interacting-BRST bridge also remains independently fail-closed.